

Brain-inspired synaptic transistors for in-situ spiking reinforcement learning with eligibility trace

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Brain-inspired reinforcement learning is pivotal for artificial general intelligence, yet current artificial neural network-based hardware lacks critical biological mechanisms like third-terminal modulated eligibility traces and dynamic reward signaling. Emerging materials address these challenges by efficiently mimicking complex reinforcement learning dynamics. Here, we demonstrate a brain-inspired spiking neural network-based reinforcement learning computing architecture using α -In₂Se₃ ferroelectric semiconductor field-effect transistor. By leveraging the intrinsic in-plane and out-of-plane polarization coupling of α -In₂Se₃, the multi-terminal conductance modulation in the device enables reward signal modulation of reinforcement learning. The ferroelectric relaxation is utilized to implement biological eligibility trace decay, thereby enhancing the algorithm's processing capability. autonomous driving tasks are then demonstrated with an RL neural network constructed by the α -In₂Se₃ transistor array, where in-situ reward-based weight updates and eligibility trace decay are performed without any external memory or computing units. Our solution enables a fully functional, energy-efficient, and low-overhead spiking-based reinforcement learning architecture.

Reinforcement learning (RL) originates from brain models in the study of biological neural systems^{1–5}, which has the potential for achieving artificial general intelligence^{6–11}. Researchers have demonstrated hardware implementations of RL based on emerging nonvolatile memory technologies (NVMs)^{12–14}. However, these artificial neural networks (ANNs) based RL implementations only implement a single aspect of RL functionality (such as weight updates¹⁵, feedback reward modulation¹³, or eligibility traces^{14,16}) rather than the complete set, with the remaining functionalities requiring additional analog circuit computations. Therefore, novel approaches are needed to achieve complete RL functionality based on NVM, enabling efficient computation.

In contrast, brain-like spiking neural networks (SNNs) offer advantages in power consumption, processing speed, and sparse computation^{17–19}, addressing the challenges faced by hardware acceleration for RL. Reward-modulated spike-timing-dependent plasticity (R-

STDP) is a brain-inspired RL algorithm designed for SNN²⁰, integrating dopamine modulation and eligibility trace mechanisms from biological systems (Fig. 1a–1c)^{3,4}. Compared to conventional STDP, feedback signals in R-STDP enhance its adaptive optimization, improving performance in learning task^{3,20}. Therefore, biologically inspired SNN-based RL holds promise for more efficient hardware implementation.

However, SNN-based RL algorithms like R-STDP usually require three hardware terminals to process reward signals (Fig. 1c)^{15,21}, two terminals serve as the pre-synaptic and post-synaptic ports, while the third terminal is used for reward modulation, which exceeds the capabilities of most two-terminal NVM devices. Additionally, the eligibility trace (ET) in RL ensures appropriate future weight updates by decaying over time, thereby helping to achieve the optimal policy^{3,4,20}. Implementing ET's exponential decay using conventional complementary metal-oxide-semiconductor (CMOS) technology

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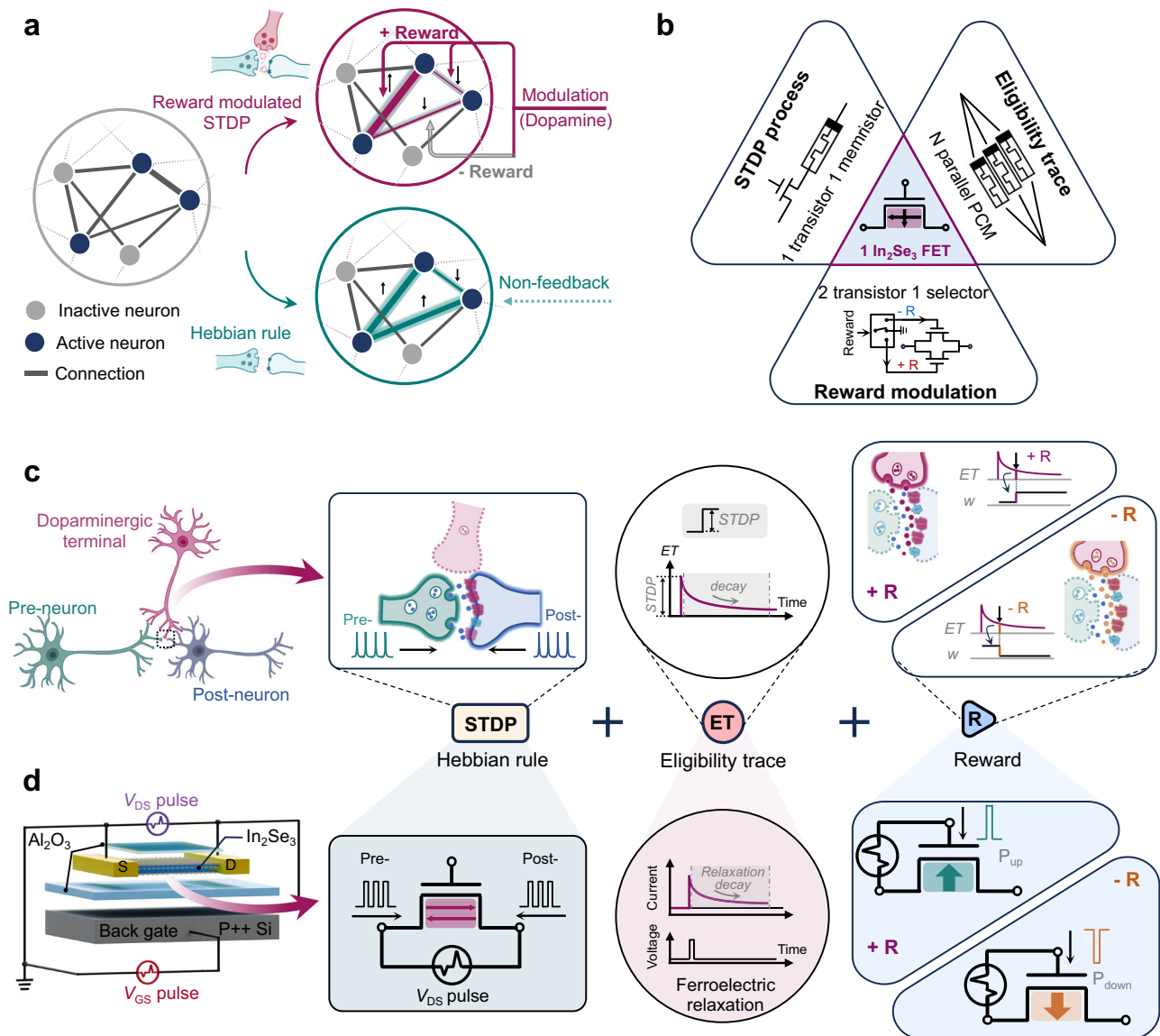


Fig. 1 | Brain-inspired in-situ spiking reinforcement learning with eligibility trace. **a** Comparison between Hebbian rule (conventional STDP) and R-STDP in biological systems. Hebbian learning relies solely on mutual stimulation for strengthening or weakening neuron connections, whereas R-STDP incorporates modulations by environmental feedback. **b** Comparison of existing solutions with the α -In₂Se₃-based hardware implementation of R-STDP. The three components of R-STDP computation (STDP, eligibility trace, and reward modulation) each require independent hardware systems in existing solutions, whereas this approach integrates all three computations within a single α -In₂Se₃ FeS-FET. **c** Detailed biological

R-STDP synaptic dynamics. (1) Pre- and post-neurons (dark green and blue color) interact for STDP behavior; (2) the STDP-stimulated weight adjustments are stored as a decaying eligibility trace; (3) a reward signal (dopamine) generated by the environment propagates back through the third terminal (pink color), arrives at the synapse and reinforces the trace, leading to weight potentiation or depression based on positive and negative reward signals. **d** These synaptic behaviors are implemented using α -In₂Se₃ FeS-FET, where in-plane and out-of-plane polarization modulations correspond to STDP and reward effects, and ferroelectric polarization relaxation mimics the eligibility trace decay.

requires analog circuits with dozens of components^{22–24}. NVM-based solutions like phase change memory (PCM) conductance drift need multiple parallel PCMs and additional synaptic weight units (Fig. 1b)^{14,16}. Therefore, more efficient device implementations for SNN-based RL are urgently called for.

In this work, we demonstrate a hardware implementation of SNN-based RL algorithm with α -In₂Se₃ ferroelectric semiconductor field-effect transistor (FeS-FET). We utilize the intrinsic physical mechanisms of the device to emulate brain-like complex analog computations, without requiring additional computational units. The brain-inspired R-STDP algorithm is mapped onto the FeS-FET for in-situ weight updates in three steps: (1) STDP Process. Pre- and post-signals are applied through the source and drain, forming V_{DS} pulses that modulate conductance by in-plane (IP) polarization; (2) Eligibility trace decay. Conductance decays

exponentially due to ferroelectric relaxation caused by depolarization field, mapping to the ET decay in the algorithm; (3) Reward modulation. During relaxation, feedback reward signals applied through the gates act as V_{GS} pulses that further modulate conductance by out-of-plane (OOP) polarization. Our approach achieves this three-step computation in a single device, mimicking the three-terminal synaptic behavior modulated by dopamine in biological systems. This work provides the complete hardware solution for in-situ weight updates of SNN-based RL, paving the way for hardware acceleration in RL.

Results

Brain-inspired synapse for spiking reinforcement learning

Figure 1a shows a comparison of the Hebbian rule (STDP) and R-STDP in the brain's neural system. Unlike STDP, which learns from the timing of

spikes alone, R-STDP adds feedback signals, aligning with RL principles. Compared to conventional STDP, the presence of positive and negative feedback signals enables R-STDP to possess adaptive optimization capabilities characteristic of RL, thereby achieving superior performance in learning tasks. Figure 1b illustrates the hardware efficiency advantages of α -In₂Se₃ FeS-FET in implementing R-STDP. Figure 1c illustrates the detailed dynamic evolution of R-STDP synapses in biological systems. Pre- and post-neurons initially generate a standard STDP effect through spike timing, but the synaptic weight does not change immediately; instead, the magnitude of the weight change is stored as an eligibility trace. The eligibility trace then decays exponentially over time, representing the strength of the learning effect. Finally, the reward signal from the third modulatory neuron is feedback, resulting in the final weight update based on the remaining magnitude of the eligibility trace. The reward signal is determined by the outcome caused by the preceding STDP event, which can be either positive or negative (Fig. 1a). The shorter the delay of the reward signal, the less the eligibility trace decays, leading to a greater amount of weight adjustment. This mechanism aligns with the principle of causality: the shorter the delay, the stronger the relationship between the event and the outcome, thereby justifying a more significant impact on weight updates^{34,20}.

Figure 1d illustrates the mechanism of implementing the R-STDP synaptic dynamics with the α -In₂Se₃ FeS-FET. By utilizing the unique IP (for STDP and ET) and OOP (for reward signal) ferroelectric polarization coupling characteristics of α -In₂Se₃, the STDP and reward parts of the algorithm can be achieved by modulating IP and OOP polarization, respectively. Furthermore, the algorithm's eligibility trace decay function is mapped to the ferroelectric polarization relaxation effect caused by the depolarization field. This approach enables in-situ spiking RL weight updates within a single transistor. Compared to existing hardware implementations, this design is simpler and achieves the complete algorithmic process computation within a single device.

Multi-terminal programmable α -In₂Se₃ synapses

The two-dimensional (2D) ferroelectric semiconductor α -In₂Se₃, with IP and OOP polarization coupling^{25–33}, enables complex analog calculations^{34–40}. Due to the coexistence of ferroelectric and resistive switching properties, α -In₂Se₃ exhibits distinct advantages and unique application scenarios compared to traditional ferroelectric and resistive switching materials (Supplementary Table 3 and 4). Furthermore, the trade-off between its ferroelectric and resistive characteristics further expands its application prospects (Supplementary Note 3). Compared to other non-volatile memory devices (e.g., FeFET and charge-trapping memories), the three-terminal modulation capability of the FeS-FET provides an additional independent control terminal. This feature can be utilized to emulate more complex biological synaptic characteristics, such as the dopamine-synapse-based R-STDP characteristics demonstrated in this work. Due to the simultaneous semiconductor and ferroelectric properties of α -In₂Se₃, FeS-FET using α -In₂Se₃ as the channel material does not require a ferroelectric gate dielectric, resulting in a simpler fabrication structure³². Figure 2a shows the cross-sectional schematic device structure of the FeS-FET. The detailed device fabrication process is described in the Methods section. Supplementary Fig. 1 illustrates the Raman spectrum of α -In₂Se₃ flake on a SiO₂ substrate. The piezoresponse force microscopy (PFM) measurement of α -In₂Se₃ (Supplementary Fig. 2) demonstrates the intrinsic coupling of IP and OOP polarization.

By applying vertical and horizontal electric fields to control the IP and OOP polarization, respectively, the I_{DS} scanning curve exhibits a significant hysteresis window (Fig. 2b and 2c), demonstrating the capability for non-volatile conductance modulation. Figure 2c shows the clockwise hysteresis consistent with the high effective oxide thickness (EOT), the EOT of 30 nm Al₂O₃ is 13 nm) condition of the gate dielectric thickness³². To rule out the possibility of clockwise hysteresis caused by charge trapping^{41,42}, Supplementary Fig. 3 presents

measurements down to 60 K. The minimal change in the hysteresis window suggests that ferroelectric polarization is the dominant factor. High-temperature characterization (Supplementary Fig. 3) also confirms the device's capability to operate in elevated temperature environments. As for the resistive switching in Fig. 2b, it is attributed to the changes in the OOP polarization coupled with the IP polarization, which alters the band bending between the α -In₂Se₃ channel and the Au- α -In₂Se₃ interface^{36,43} (Supplementary Fig. 4).

The polarization modulation schematics for IP (Fig. 2d–2f) and OOP (Fig. 2i) are shown in Fig. 2. These two modulation methods correspond to STDP and reward signals in biological systems, respectively. Figure 2e and 2h present the analog modulation of channel conductance under V_{DS} and V_{GS} pulses, respectively. The large hysteresis window under V_{GS} (± 5 V, ON/OFF ratio $> 10^5$) contributes to excellent conductance modulation range ($G_{max}/G_{min} > 25$) and multiple conductance states (> 32). The V_{DS} pulses can also produce analog modulation on channel conductance ($G_{max}/G_{min} > 6$, > 32 states). The results of 500 cycling measurement and the analog modulation under different pulse widths/amplitudes are shown in Supplementary Fig. 5 and Supplementary Fig. 6, demonstrating excellent cycling consistency. Because the thickness (~ 40 nm) of α -In₂Se₃ is much smaller than the channel length (1 μ m), the displacement field of V_{GS} (5 V under 30 nm gate insulator is $170 \text{ V } \mu\text{m}^{-1}$) is much higher than that of V_{DS} (7 V under 1 μ m channel length is $7 \text{ V } \mu\text{m}^{-1}$), which leads to stronger modulation of V_{GS} . This characteristic aligns well with the requirements of the R-STDP algorithm in biological systems, where STDP signal (V_{DS}) modulation is weaker than reward signal (V_{GS}) modulation^{3,4}. This compatibility facilitates our subsequent application of FeS-FET for R-STDP synapses.

The excitatory postsynaptic current (EPSC) and inhibitory postsynaptic current (IPSC) under two consecutive V_{DS} pulses (± 9 V, 20 ms. The read voltage is 0.1 V) is shown in Fig. 2f. The application of a positive V_{DS} pulse is equivalent to an increase in the read voltage, and together with the change in ferroelectric polarization, results in a rapid increase in current. Polarization relaxation causes the current to partially decrease after the pulse is removed, while residual polarization results in a non-volatile increase of conductance. The situation is similar in the case of a negative pulse.

Figure 2i shows the current response under two consecutive V_{GS} pulses (7 V, 15 ms. The read voltage is 0.1 V). Upon application of a positive voltage pulse, the current increases sharply due to the attraction of electrons into the channel. After the pulse is removed, the current decreases rapidly as ferroelectric polarization dominates. Simultaneously, the presence of the depolarization field causes partial polarization relaxation⁴⁴, leading to a reduction in residual polarization and a subsequent increase in current. The positive V_{GS} pulses result in IPSC, and the negative V_{GS} pulses result in EPSC. The current relaxation caused by polarization relaxation in α -In₂Se₃ FeS-FET is consistent with the eligibility trace decay in RL algorithms, thereby providing an efficient hardware implementation scheme that does not require additional circuitry. Supplementary Fig. 7 illustrates the conductance retention under V_{GS} pulses.

In addition, Supplementary Fig. 8 shows the standard STDP function via V_{DS} pulses and the pulse scheme used for STDP measurement. Unlike existing research, where waveforms are pre-computed and then input to the transistor gate to achieve STDP^{15,31,35}, this approach naturally realizes STDP by using the source and drain as the pre- and post-synaptic terminals, respectively, allowing STDP to occur through the superposition of pre- and post-neuron pulses. The diverse synaptic behaviors of α -In₂Se₃ FeS-FET illustrate its applicability as an artificial synapse in spiking-based reinforcement learning.

Reward-modulated synapses with eligibility trace

In biological R-STDP systems, reward signal (dopamine) and eligibility traces play a critical role^{3,4}. Figure 3a shows that the eligibility trace

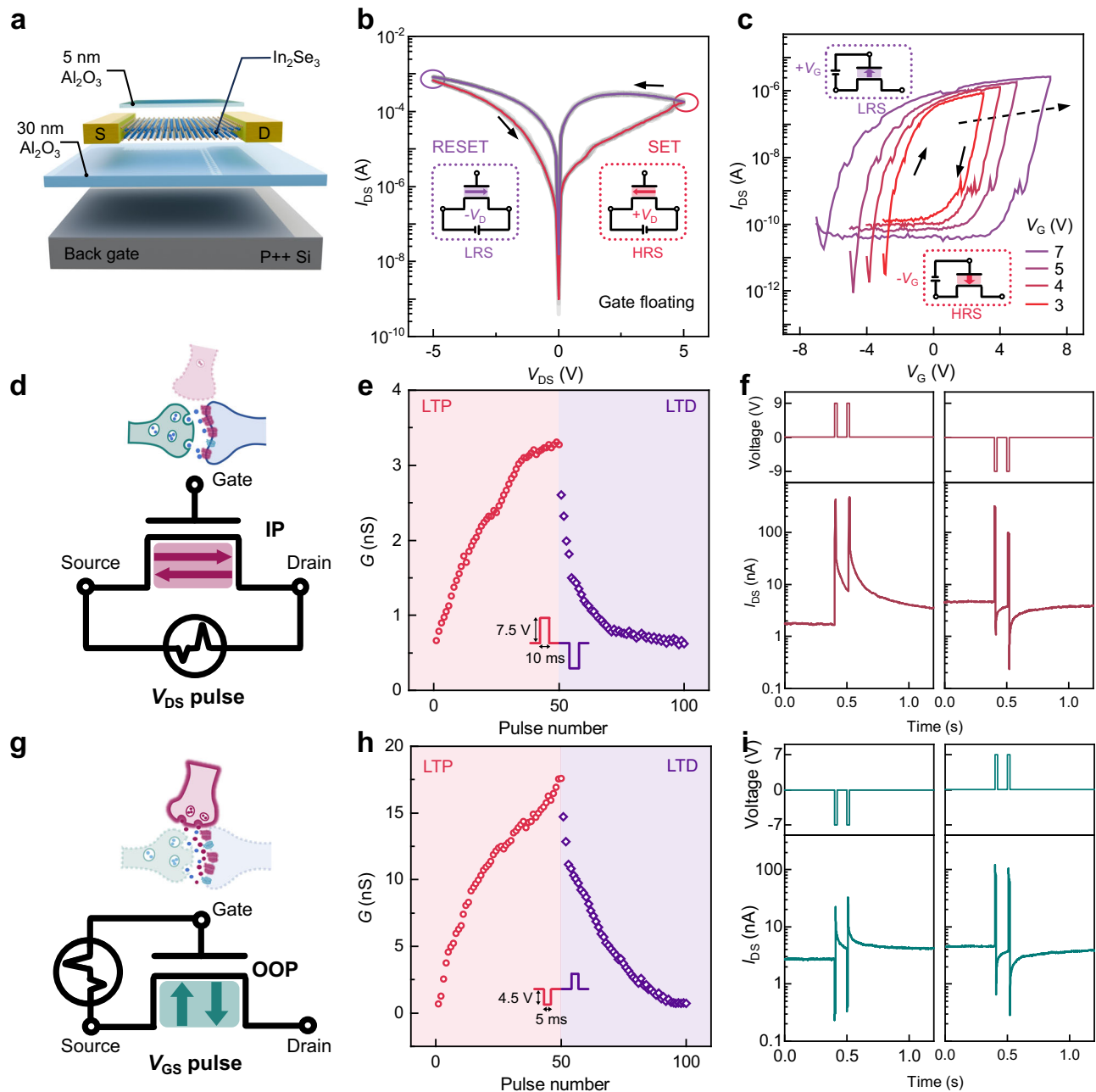


Fig. 2 | Multi-terminal programmable synaptic behavior of α -In₂Se₃ FeS-FET. **a** Schematic structure of the α -In₂Se₃ device. **b, c**, drain current I_{DS} as a function of source-drain voltages (**b**) and gate voltages (**c**). In **b**, the shaded area represents the results of ten consecutive tests. The hysteresis window is caused by IP (**b**) and OOP (**c**) polarization, respectively. The schematic diagram of IP (**d**) and OOP (**e**) pulses

modulation. The analog modulation behaviors of conductance G under V_{DS} (**e**) and V_{GS} (**h**) pulses. EPSC and IPSC under two consecutive V_{DS} (**f**) and V_{GS} (**i**) pulses. HRS: High resistance state, LRS: Low resistance state, LTP: Long-term potentiation, LTD: Long-term depression.

captures the strength of the STDP effect and decays exponentially over time. When a reward signal arrives, the weight update is based on the remaining trace. This process aligns with the Markov chain in reinforcement learning⁴⁵, where past and future events together influence weight updates. In R-STDP, the earlier the reward, the stronger the link between the STDP event and the reward, leading to a larger weight adjustment due to less decay in the eligibility trace (Fig. 3a). The simplified R-STDP rule is as follows³: $\Delta w = \text{STDP}(t) \cdot E(t) \cdot R(t)$, Δw is the weight change, $E(t)$ and $R(t)$ represent the eligibility trace and reward signal, respectively (Fig. 3a). The schematic diagram of using α -In₂Se₃ FeS-FET to implement R-STDP in-situ update is shown in Fig. 3a: (1) V_{DS} represents the effective pulse composed of pre- and post-synaptic

pulses, indicating the STDP signal; (2) The relaxation effect of the conductance after the pulse represents the decay of the eligibility trace; (3) The delayed V_{GS} pulse represents the delayed reward signal, a negative V_{GS} pulse (resulting in increased conductance) corresponds to a positive reward, whereas a positive V_{GS} pulse (resulting in decreased conductance) corresponds to a negative reward. The combined effect of V_{DS} and V_{GS} results in the final conductance update.

The R-STDP update formula requires that the weight not update when only one signal is present. The weight updates only when both STDP(t) and $R(t)$ signals are present simultaneously (Supplementary Fig. 9). This requirement can be met by setting the amplitude of a single pulse near the ferroelectric threshold. The threshold test results

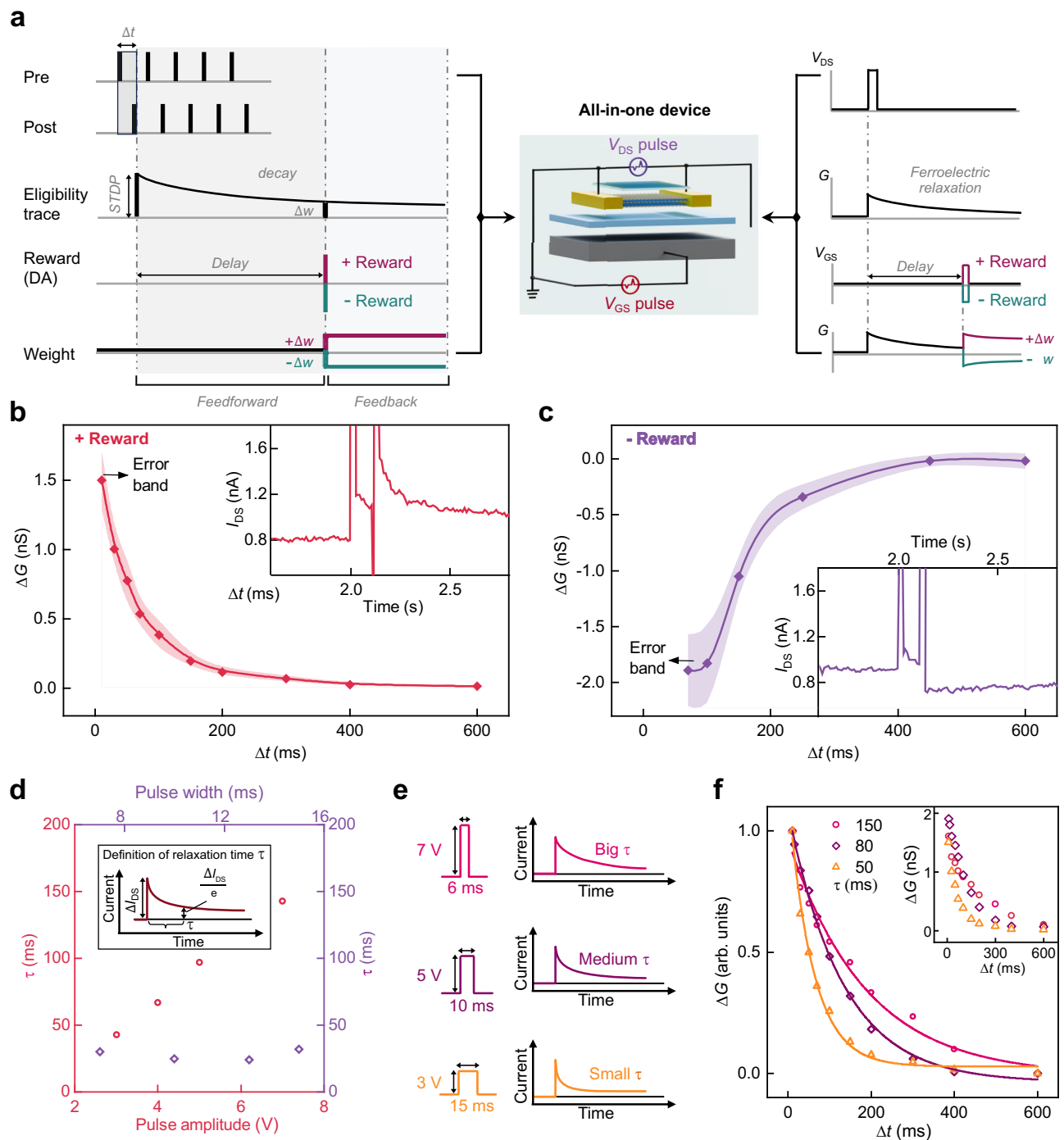


Fig. 3 | Reward-modulated FeS-FET synapses with eligibility trace.

a, b Schematic of biological R-STDP (left) and R-STDP via α - In_2Se_3 FeS-FET (right). R-STDP statistical curves under positive (**b**) and negative (**c**) reward, respectively. Inset: Detailed conductance modulation, the first pulse is V_{DS} and the second pulse is V_{GS} . In (**b**) and (**c**), the error bands represent the standard error. **d** Relaxation time

under different V_{DS} pulse heights and widths. Inset: Definition of relaxation time constant. **e** Three different V_{DS} pulse waveforms are used to achieve three distinct relaxation time constants. **f** R-STDP curves under three different relaxation time constants of **e**. Inset: Unnormalized raw data. All with read voltages (**b–f**) 0.1 V.

for V_{DS} and V_{GS} are shown in Supplementary Fig. 10, with values of 5 V and 2.1 V. More importantly, in R-STDP, the longer the delay of the reward signal, the more the eligibility trace decays, and the smaller the final weight update value. The α - In_2Se_3 FeS-FET achieves this through the time interval between V_{DS} and V_{GS} pulses. The larger the time interval, the more pronounced the ferroelectric relaxation, resulting in a weaker modulation. Supplementary Fig. 11 shows the modulation results for different time intervals, which align with the expectations. Statistical results (Fig. 3b and 3c) indicate that the greater the delay of

the reward signal, the weaker the weight modulation. The results align well with the standard R-STDP curve. This demonstrates that α - In_2Se_3 FeS-FET can realize the functionality of R-STDP synapses.

As a widely used concept in RL, eligibility trace can effectively improve learning efficiency and has been validated in biological systems^{4,46–48}. However, on CMOS-based or NVMs-based computing platforms^{14,16,24,49}, implementing decaying eligibility traces exponentially requires significant hardware overhead (For example, in Fig. 1b, multiple PCMs in parallel are required to utilize their resistance drift to achieve

the eligibility trace). Furthermore, it is also difficult for FeFET and charge-trapping memories to realize a decaying trace via simple pulse modulation. While often considered a non-ideal effect, ferroelectric polarization relaxation has been successfully exploited in recent studies to enable complex functionalities such as reservoir computing^{36,37} and associative learning³⁰. Rather than a detriment, relaxation serves as a valuable, controllable mechanism^{36,51} for neuromorphic and unconventional computing. In this work, we leverage ferroelectric relaxation to naturally achieve eligibility trace decay with a tunable relaxation time.

The definition of the relaxation time τ is shown in Fig. 3d. To achieve tunable τ , measurements with varying V_{DS} pulse amplitudes and widths are performed (Supplementary Fig. 12). Figure 3d indicates that the larger the amplitude (3–7 V), the greater the τ , while the pulse width (7–15 ms) has no effect on τ . The differences observed in the hysteresis loops under different V_{DS} demonstrate the influence of V_{DS} on polarization and relaxation (Supplementary Fig. 13). This may be related to the sensitivity of the depolarization field to voltage⁵¹. Therefore, by increasing the amplitude and reducing the pulse width (Fig. 3e), we can achieve different eligibility trace relaxation times without altering the conductance with a single V_{DS} pulse, in accordance with the requirements of the R-STDP formula. Figure 3f shows the R-STDP test results under three different τ values. The tunable range based on α -In₂Se₃ FeS-FET can reach up to 28–189 ms, which is comparable to the eligibility trace times in biological systems⁴. Tunable relaxation time τ of eligibility trace by electrical pulses allows the algorithm to achieve optimal performance across various scenarios, particularly in complex problems^{20,47}.

Supplementary Table 1 summarizes the comparison of R-STDP implementation by different hardware. Compared to systems that incur higher hardware costs and only partially implement R-STDP functions (synaptic weights, eligibility traces, and reward signals), this work integrates all R-STDP functionalities into a single α -In₂Se₃ FeS-FET, enabling in-situ weight updates and incorporating tunable eligibility traces. Our approach reduces the hardware complexity by a factor of several times. This learning rule is realized by our FeS-FET with about 10 μm^2 footprint. Furthermore, it exhibits low operating energy consumption, approximately 65 pJ per R-STDP event (Supplementary Fig. 14), demonstrating a significant advantage over comparable devices (Supplementary Table 1). Therefore, it indicates the significant predominance in area, energy, and latency of the hardware SNN system for RL. Meanwhile, the power consumption of the printed circuit board (PCB) control circuit is significantly higher than that of the device itself (Supplementary Note 1). Future improvements are expected to achieve much higher energy efficiency through custom on-chip integration^{52,53} or novel memristor-based ADC⁵⁴. Supplementary Table 2 summarizes α -In₂Se₃-based synaptic devices. The R-STDP synapse presented in this work exhibits distinct advantages in terms of joint IP and OOP modulation, conductance range, and energy consumption.

Spiking reinforcement learning neural network

RL has gained widespread application in areas such as autonomous driving, gaming, and robotic control due to its ability to handle open-ended tasks effectively. To demonstrate the application of In₂Se₃ FeS-FET-based R-STDP synaptic reinforcement learning, we set up an autonomous driving scenario as shown in Fig. 4a. Supplementary Fig. 15 illustrates the detailed simulation scenario. The lane scene captured by the vehicle's camera is compressed into a 3×6 pixels format⁵⁵, where each pixel is encoded by pulse count to generate 18 pulse sequences that are separately input to 18 input neurons of the spiking RL neural network connected by R-STDP synapses.

A 6×8 In₂Se₃ FeS-FET array and a corresponding PCB circuit operating system are fabricated to accomplish the autonomous driving task (Fig. 4b and Supplementary Fig. 16). The measured pulse output from the PCB writing circuitry in Supplementary Fig. 17 demonstrates that the circuit system exhibits extremely low noise. An

analysis of the circuit system's power consumption is presented in Supplementary Note 1. Figure 4c presents the R-STDP curve of the 36 devices in the array, with detailed curves presented in Supplementary Fig. 18. The I_D - V_G (OOP polarization) and I_D - V_D (IP polarization) curves also demonstrate good consistency (Supplementary Fig. 19 and Supplementary Fig. 20). Supplementary Fig. 21 presents the statistical information of the on-off ratio. The process of writing weights to three individual devices via the PCB circuit system, along with the corresponding writing results, is illustrated in Supplementary Fig. 22. The weight distribution pre-trained by the software is shown in Supplementary Fig. 23. The actual weight distribution, written into the array according to the required weights based on the R-STDP curve, is shown in Fig. 4d. Figure 4e presents that the performance of the experimental array with written weights is largely consistent with the performance of the weights pre-trained by software in the autonomous driving task. Supplementary Fig. 24 shows the weight distribution at 300 seconds and 800 seconds after array writing, along with the corresponding performance. Even in the presence of a certain degree of Device-to-Device (D2D) variation or noise variation, the network remains capable of successful convergence, exhibiting a slight reduction in convergence speed (Supplementary Figs. 25 and 26). This demonstrates the strong redundancy capability of the In₂Se₃ FeS-FET-based RL neural network. This is attributed to the closed-loop nature of in-situ training, where the iterative weight updates dynamically compensate for static device errors. Consequently, the algorithm automatically adjusts the number of update steps to reach the target conductance, effectively mitigating the impact of individual device variability^{56–61}.

As mentioned earlier, eligibility trace is a key concept in RL and can significantly improve algorithm performance. The adjustable eligibility trace decay constant ensures optimal performance across different tasks. In₂Se₃-based synapses can easily adjust the decay constant through different pulses (Fig. 3e, f). To validate the effect of eligibility trace, Fig. 4f shows the performance of the training process with different decay time constants. The details of weight evolution during training are shown in Supplementary Fig. 27. The results indicate that as the decay time constant increases, the convergence speed initially rises and then falls, reaching its peak at 130 ms. Additionally, the convergence speed decreases without eligibility traces. More data on convergence speed and time constants are presented in the inset of Fig. 4f. This result indicates that an adjustable eligibility trace time constant significantly impacts algorithm performance. R-STDP synapses based on In₂Se₃ FeS-FET effectively enhance the algorithm's processing capability through relaxation time constants (28–189 ms) adjustable by pulse amplitude.

Discussion

We present the In₂Se₃ FeS-FET array that can be used for in-situ updates in spiking reinforcement learning. First, by simultaneously utilizing the OOP and IP polarization of In₂Se₃, the FeS-FET exhibits three-terminal control, enabling the reward feedback function in RL. The complex calculation of the three-terminal signals in RL is completed in-situ through the flipping of ferroelectric polarization. Second, the eligibility trace, as a crucial parameter determining the performance of R-STDP here, is emulated through the relaxation of ferroelectric polarization. Moreover, the tunability of eligibility trace is realized through exploiting the dependence of ferroelectric relaxation process on the programming voltages, and the impact of its optimization on RL tasks is further illustrated in a lane-keeping autonomous driving application. Our RL hardware implementation provides an efficient solution for future edge computing.

Methods

Device fabrication

The transistor was fabricated on Al₂O₃/p-Si (30 nm/500 μm) substrate by transferring α -In₂Se₃ thin flakes using polydimethylsiloxane-assisted exfoliation and transfer methods. Then the source and drain

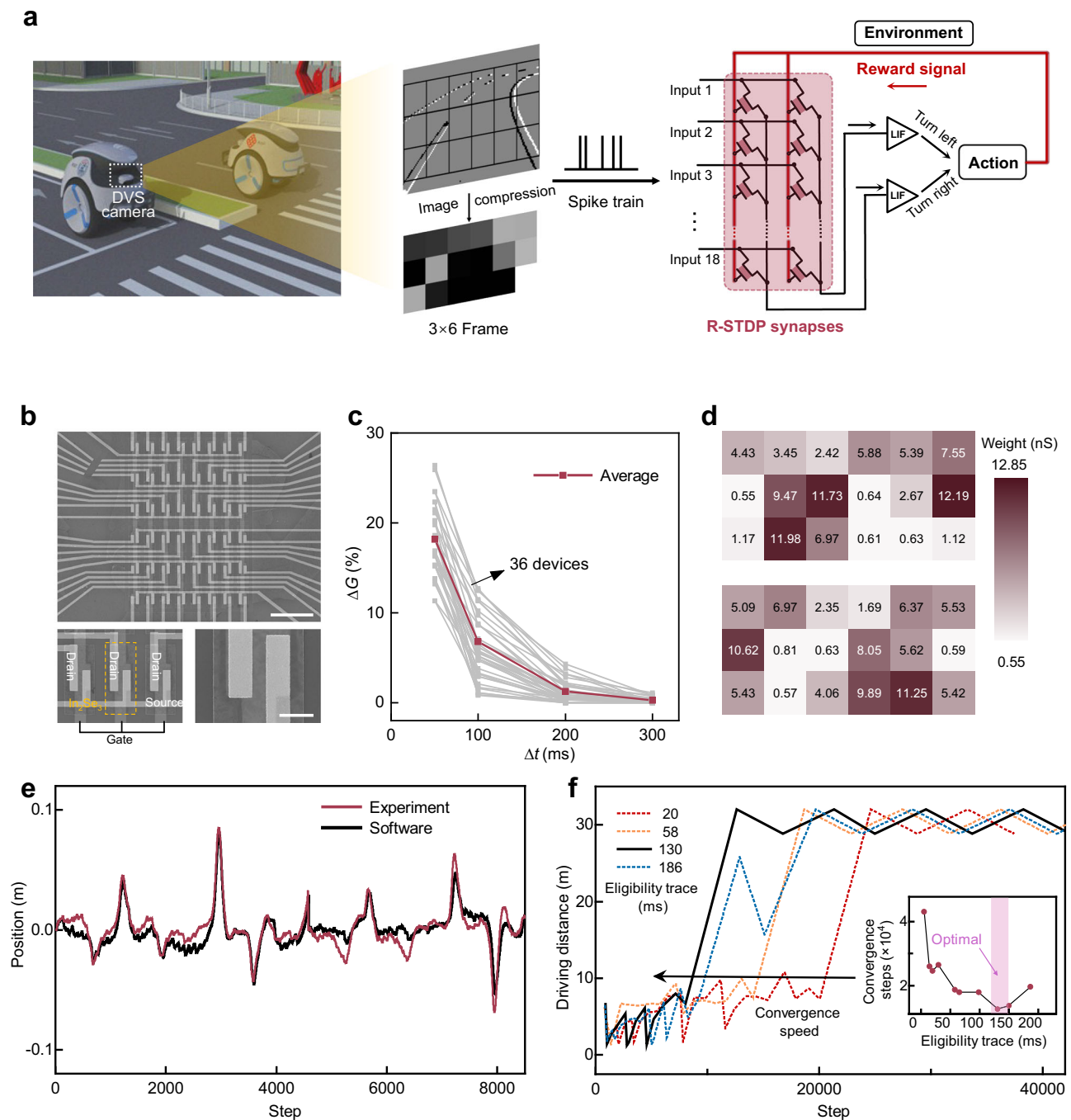


Fig. 4 | In_2Se_3 FeS-FET array for spiking RL neural network and application in lane keeping task of autonomous driving. **a** The schematic diagram of an autonomous driving scenario. The input image is compressed to a 3×6 format, which is encoded by pulse count and input to 18 input neurons. Two output neurons control the left and right directions, respectively. The neurons are connected through R-STDP synapses. **b** The scanning electron microscope (SEM) image of the

6×8 array. The top scale is $20 \mu\text{m}$, and the bottom scale is $2 \mu\text{m}$. **c** The R-STDP curves of the 36 devices in the array. **d** The actual weight distribution of the array from the experiment. **e** Comparison of performance between software-pretrained and array-written weights in the autonomous driving scenario. The x-axis is the step size, and the y-axis is the distance from the lane center. **f** Performance under different decay constants of eligibility trace. Inset: The optimal τ is about 120–150 ms.

electrode regions (with $1 \mu\text{m}$ between them) were patterned by electron beam lithography. Cr/Au (10/50 nm) electrodes were deposited through e-beam evaporation followed by a lift-off process. Finally, 5 nm of Al_2O_3 was covered by atomic layer deposition technology as a passivation layer.

As for the $\alpha\text{-In}_2\text{Se}_3$ FeS-FET array, the local bottom gate electrodes (Cr/Au, 5 nm/25 nm) were patterned by electron beam lithography on $\text{SiO}_2/\text{p-Si}$ (300 nm/500 μm) substrate. Then, 30 nm Al_2O_3 is deposited as the gate insulation layer by atomic layer deposition. The $\alpha\text{-In}_2\text{Se}_3$

synthesized by Flux-assisted growth is transferred to the gate electrode via polyvinyl alcohol assisted transfer. The $\alpha\text{-In}_2\text{Se}_3$ is patterned by electron beam lithography, then the channel region is formed by inductively coupled plasma etching ($\text{Cl}_2 + \text{SF}_6$). Finally, top source and drain electrodes (Cr/Au, 10 nm/50 nm) were patterned.

Device characterization

Room-temperature electrical measurements were carried out using a Lakeshore probe station equipped with an Agilent B1500A and

Table 1 | Simulation parameters specification

Steering model	Max. speed	$V_{\max} = 15$ ms
	Min. speed	$V_{\min} = 10$ ms
	Turn constant	$C_{\text{turn}} = 0.5$
	Max spikes during a simulation step	$N_{\max} = 15$
SNN simulation	Simulation time	50 ms
	Tim resolution	0.1 ms
LIF	Resting membrane potential	$E_L = 700$ mV
	Capacity of the membrane	$C_m = 2500$ pF
	Duration of refractory period	$t_{\text{ref}} = 20$ ms
	Spike threshold	$V_{\text{th}} = 550$ mV
R-STDP synapse	Amplitude of weight change for facilitation	$A_+ = 1.0$
	Amplitude of weight change for depression	$A_- = 1.0$
	STDP time constant	$\tau = 200$ ms
	Time constant of eligibility trace	$\tau_{\text{ET}} = 5 - 200$ ms
	Minimal synaptic weight	0.5 nS
	Maximal synaptic weight	12.8 nS
	Reward constant	$c_r = 0.1$

B1530 semiconductor parameter analyzer. The high/low-temperature measurements were carried out in a Lakeshore cryogenic probe station in vacuum conditions, equipped with an Agilent B1500A semiconductor parameter analyzer.

Raman spectra of the exfoliated $\alpha\text{-In}_2\text{Se}_3$ flakes on the SiO_2 substrate were obtained using a Raman imaging microscope (Horiba LabRAM Odyssey) with an excitation laser of 532 nm. Scanning electron microscopy (SEM) images were obtained using a field emission scanning electron microscope (JSM-7900F). PFM measurements were performed using a multimode AFM device (Bruker Dimension Icon).

Reinforcement learning based spiking neural network

The autonomous driving environment set up is simulated by Python based on the reported model⁶². We constructed a spiking neural network with a scale of 18×2 to process an autonomous driving task, where 18 input neurons represent the 3×6 input image information, and two output neurons correspond to left and right turns. The parameters of the simulation environment are listed in Table 1. The synaptic weight update behaviors are based on the R-STDP characteristics of the $\alpha\text{-In}_2\text{Se}_3$ synaptic cell. The neuron mode is based on the leaky integrate and fire neuron (LIF neuron). Based on the spiking output and the reward signal, the synaptic conductance is then updated based on the R-STDP characteristics. The specific weight update method is as follows.

The STDP as a function of the time difference between pre- and postsynaptic spikes is defined as:

$$\Delta t = t_{\text{post}} - t_{\text{pre}} \quad (1)$$

$$W(\Delta t) = \begin{cases} A_+ e^{-\frac{\Delta t}{\tau_+}}, & \Delta t \geq 0 \\ -A_- e^{-\frac{\Delta t}{\tau_-}}, & \Delta t < 0 \end{cases} \quad (2)$$

$$\Delta w = \sum_{t_{\text{pre}}} \sum_{t_{\text{post}}} W(\Delta t) \quad (3)$$

where w is the synaptic weight. Δw is the change of the synaptic weight. t_{pre} and t_{post} stand for the timing of the firing spike from pre-neuron and post-neuron. A_+ and A_- represent positive constants scaling the strength of potentiation and depression, respectively. τ_+

and τ_- are positive time constants defining the width of the positive and negative learning window.

In the R-STDP, the synaptic weight w changes with the reward signal R . The eligibility trace of a synapse can be defined as:

$$\dot{c}(t) = -\frac{c}{\tau_c} + W(\Delta t)\delta(t - s_{\text{pre/post}})C_1 \quad (4)$$

where c is an eligibility trace. $s_{\text{pre/post}}$ means the time of a pre or postsynaptic spikes. C_1 is a constant coefficient. τ_c is a time constant of the eligibility trace. δ is the Dirac delta function.

$$\dot{w}(t) = R(t) \times c(t) \quad (5)$$

where $R(t)$ is the reward signal. More details on the R-STDP mechanism can be found in the paper^{3,63}.

Data availability

The data that support the plots within this paper and other findings of this study are available from the corresponding author upon request. Source data are provided with this paper.

Code availability

The code for the SNN with the scheme is available from the corresponding author with detailed explanations upon request.

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Author contributions

Y.H. and Y.C. conceived and supervised the project. Y.W. designed the experiment. Y.W., W.X., and J.Y. fabricated the devices. Y.W. and W.X. performed the Raman and Atomic Force Microscope characterizations. Y.W. performed the device measurement. W.X., Y.H. and Y.W. designed and performed the neural network simulations. Y.W., W.X., Y.Z., C.Z., and X.M. analysed the data. Y.W. and Y.C. wrote the paper. All the authors discussed the results and implications and reviewed the paper.

Competing interests

The authors declare no competing interests.

Additional information

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